

6.4.9 Software Operation Process

This International Standard specializes the Software Operation Process, a specialization of the **Operation Process** in ISO/IEC 15288. Users may claim conformance to either the ISO/IEC 15288 process or this subclause (6.4.9).

6.4.9.1 Purpose

The purpose of the Software Operation Process is to operate the software product in its intended environment and provide customer support throughout its operational life.

6.4.9.2 Outcomes

As a result of successful implementation of the Software Operation Process:

- a) An operation strategy is defined.
 - b) Conditions for correct software operation in its intended environment are identified and evaluated.
 - c) Software operation in its intended environment is tested and determined.
 - d) The software is operated in its intended environment under defined conditions.
 - e) Assistance and consultation are provided to customers in accordance with agreements.
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6.4.9.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.9.3.1 Preparation for Operation

This activity establishes the plans, procedures, and environment for software operation.

- 6.4.9.3.1.1 Develop and document an operation plan that defines operational standards, responsibilities, procedures, and resources for performing operational tasks.

- 6.4.9.3.1.2 Establish procedures for receiving, recording, resolving, tracking, and providing feedback on problems. Problems shall be recorded and entered into the **Software Problem Resolution Process** (7.2.8).
 - 6.4.9.3.1.3 Establish procedures for testing the software in its operational environment prior to release for operational use. Problem reports and modification requests shall be entered into the **Software Maintenance Process** (6.4.10).
 - 6.4.9.3.1.4 Define procedures for releasing the software product for operational use once operational testing criteria are satisfied.
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6.4.9.3.2 Operational Use

This activity executes the software in its intended environment and manages transitions between releases or systems.

- 6.4.9.3.2.1 For each software product release, perform operational testing and release the product for operational use if the specified operational criteria are met.
 - 6.4.9.3.2.2 Initialize, execute, and operate the software code and databases according to their intended purpose.
 - 6.4.9.3.2.3 Activate the software in its intended operational situation to deliver instances of service or continuous service.
 - 6.4.9.3.2.4 Where agreed, maintain continuous service capacity and quality when replacing an existing system being retired. During any specified changeover or concurrent operation period, manage the transfer of services to ensure stakeholder needs continue to be met.
 - 6.4.9.3.2.5 Terminate the system according to the operation plan when required (e.g., during replacement, decommissioning, or upgrades).
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6.4.9.3.3 Monitoring and Evaluation

This activity ensures that operational performance is monitored and risks are identified.

- 6.4.9.3.3.1 Operate the system in its intended environment with the required resources and under specified conditions.
 - 6.4.9.3.3.2 Identify and monitor risks to product operation on an ongoing basis.
 - 6.4.9.3.3.3 Monitor operational service regularly against defined criteria, performance objectives, and service-level agreements.
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6.4.9.3.4 Customer Support

This activity provides assistance and consultation to customers and users.

- 6.4.9.3.4.1 Provide assistance and consultation to users as requested, including training, documentation, and other support services necessary for effective product use.
- 6.4.9.3.4.2 Record and monitor user requests and actions taken in accordance with organizational procedures.
- 6.4.9.3.4.3 Forward user requests that require corrective action to the **Software Maintenance Process** (6.4.10) for resolution. Inform originators of planned and completed actions.
- 6.4.9.3.4.4 Forward identified operational problems to the **Software Problem Resolution Process** (7.2.8) for handling.
- 6.4.9.3.4.5 When a temporary workaround exists for a reported problem, communicate this to the originator. Apply permanent corrections, previously omitted functions or features, and system improvements to the operational software product through the **Software Maintenance Process** (6.4.10).

NOTE: Assistance may include training, operational documentation, or helpdesk services.

Cross-References

This process interacts with the following:

- **Software Problem Resolution Process (7.2.8)** — for tracking and resolving operational issues.
- **Software Maintenance Process (6.4.10)** — for applying fixes, enhancements, and updates.
- **Software Installation Process (6.4.7)** — for deploying software into the operational environment prior to operation.
- **Measurement Process (6.3.2)** — for tracking operational performance and service levels.